

When Mamba Needs Attention

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Abstract. We ask whether a linear-time state-space (Mamba) encoder can drive routing decisions as well as a quadratic-time attention encoder on Euclidean TSP-100, under identical POMO reinforcement learning and a byte-identical decoder, with only the encoder swapped and parameter counts matched.^a The obvious recipe, a Hilbert-curve serialization scanned by a unidirectional Mamba, is killed: at a matched 250k-step budget it reaches a 7.50 ± 1.03 single-trajectory optimality gap over three seeds (8.34% on the pre-registered seed) versus attention’s 2.95%. Diagnostics localize the failure to a short, one-directional receptive field rather than a wiring bug or representation collapse, and a geometry study shows the Hilbert prior scatters $\sim 14\%$ of spatial neighbors far apart on the scan line. An ordering ablation initially suggested a strict random $<$ sort $<$ Hilbert inversion, but two further seeds refuted the strict claim; the surviving, scoped statement is that a locality prior confers no benefit for *causal* scans and roughly doubles seed variance (σ 1.03 vs 0.53). Repair follows the diagnosis: pooled global channels fail, but a param-matched bidirectional scan reaches multistart parity with attention (1.94 ± 0.09 vs 1.91) and cuts seed variance tenfold, and a hybrid inserting *one* attention layer into a bidirectional-Mamba stack (4.5% fewer parameters) beats attention on multistart tour quality at a converged 500k budget (worst hybrid seed 1.365 $<$ best attention 1.589). All claims are scoped to $N=100$; the $O(N)$ payoff remains untested.

^aCode and pre-registered decision rules: <https://github.com/Umaraslam66/serpentine>.

1 INTRODUCTION

Neural constructive solvers for routing problems build a tour one node at a time, and the dominant architecture is an attention encoder followed by an attention-based autoregressive decoder [Kool et al., 2019, Kwon et al., 2020]. The encoder’s job is to embed each city so that the decoder can compare candidates; attention gives every node an all-pairs view of the instance at $O(N^2)$ cost per layer. State-space models such as Mamba [Gu and Dao, 2023, Gu et al., 2022] process a sequence in $O(N)$ time with a selective linear recurrence, and have become a competitive backbone for point clouds [Liang et al., 2024, Wu et al., 2024, Zhang et al., 2025a]. This raises a concrete question for neural combinatorial optimization (NCO): *can an $O(N)$ state-space encoder route as well as an $O(N^2)$ attention encoder, under the reinforcement-learning training that NCO actually uses?*

We treat this as a fail-fast experiment. We fix the decoder, the training algorithm (POMO REINFORCE [Kwon et al., 2020, Williams, 1992]), the optimizer, the budget, the evaluation set, and the parameter count,

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and swap only the encoder. A state-space encoder needs a serialization, an order in which to scan the unordered node set, and the natural choice for 2D points is a space-filling Hilbert curve [Hilbert, 1891, Wu et al., 2024]. The candidate is therefore a Hilbert-ordered unidirectional Mamba encoder; we ask whether it clears the attention baseline.

It does not. We then use the failure as a diagnostic object rather than a dead end, and follow the mechanism to a repair. Our contributions, scoped honestly:

- **A clean kill of the vanilla recipe under RL.** At matched parameters and budget on TSP-100, a Hilbert-ordered unidirectional Mamba encoder reaches a 7.50 ± 1.03 single-trajectory gap across three seeds (its *best* seed, 6.36%, is still more than 3 points behind attention’s 2.95%; §4). The kill was pre-registered.
- **A localized diagnosis.** The failure is not a wiring bug (order demonstrably reaches the decoder) and not representation collapse (the Mamba embeddings are higher-rank than attention’s); finite-difference probes show a short, strictly causal receptive field, and a model-free geometry study shows the Hilbert prior scatters $\sim 14\%$ of spatial neighbors (§4).
- **A methodological finding about ordering priors and seeds.** A single-seed ablation suggested a strict random < sort < Hilbert inversion; two further seeds refuted the strict claim. The surviving statement is that a locality prior buys nothing for *causal* scans and doubles seed variance, and, crucially, that its value depends on scan causality: with a bidirectional scan, random ordering *hurts* (§5).
- **A diagnosis-guided repair.** Pooled global channels do not help under RL; a param-matched bidirectional scan reaches multistart parity with attention and stabilizes training tenfold; and a hybrid with a single attention layer beats attention on converged multistart tour quality with fewer parameters (§6).
- **A decode-diversity observation attention lacks.** Because a scan encoder has an ordering degree of freedom, best-of- k serialization ensembling at inference improves tour quality for free; attention has no such knob (§6).

We are careful about novelty. Bidirectional scans, space-filling serializations, and SSM–attention hybrids are all prior art in the point-cloud and language-model literatures (§2); what is new here is the controlled, matched-budget evidence for *routing decisions under policy-gradient RL at $N=100$* , and the contrast between exact sparse attention and pooled global channels as repairs.

2 RELATED WORK

Neural combinatorial optimization. Pointer networks introduced sequence-to-sequence attention for combinatorial outputs [Vinyals et al., 2015]; Bello et al. [2016] trained such models with reinforcement learning; Kool et al. [2019] established the attention-encoder / attention-decoder template for routing, and POMO [Kwon et al., 2020] added multiple-optima rollouts with a shared baseline and instance augmentation. Our training and evaluation follow POMO; classical LKH-3 [Helsgaun, 2017] provides the optimality oracle.

State-space models. Structured state-space models [Gu et al., 2022] and the selective Mamba variant [Gu and Dao, 2023] give $O(N)$ sequence modeling with a linear recurrence, and are the backbone we test as an encoder. Mamba has also been applied to RL as offline trajectory modeling [Ota, 2024], a different problem from the constructive routing we study.

Mamba for point clouds. Mamba has been adapted extensively to unordered 3D points, which, like routing, require serializing a set. The canonical observation that a unidirectional scan has a causal, all-pairs-free receptive field relative to attention is from PointMamba [Liang et al., 2024]. Known mitigations, all prior art, include bidirectional scanning [Zhu et al., 2024, Qu et al., 2025, Li et al., 2025], multi-curve or shuffled serialization [Wu et al., 2024], and hybrid local/global aggregation [Zhang et al., 2025b]. HydraMamba [Qu et al., 2025] further notes that a bidirectional scan still compresses history into a finite hidden state, which is why we expect bidirectionality to close the gap only partially. Point Cloud Mamba [Zhang et al., 2025a] reports that simple orders are roughly on par with space-filling curves. We reuse this diagnostic vocabulary; we do not claim it.

Mamba for combinatorial optimization. ECO [Xu et al., 2026] applies a Mamba encoder–decoder to TSP/CVRP, but trains via supervised fine-tuning plus iterative preference optimization (not policy-gradient RL), targets large instances (its smallest reported TSP is $N=200$), and never tests $N=100$. The cell we occupy (matched-budget POMO REINFORCE at $N=100$, decoder held fixed) is not covered by ECO, and our kill of vanilla Mamba+Hilbert at $N=100$ is consistent with ECO’s large- N focus and its reliance on supervised bootstrapping.

Hybrid SSM–attention models. Interleaving attention layers into an SSM backbone is established in language models: Jamba [Lieber et al., 2024] and Zamba [Glorioso et al., 2024] both mix Mamba and attention blocks for efficiency. Our hybrid uses the same idea; the contribution is not the interleaving pattern but the RL-routing evidence that *one* exact attention layer succeeds where pooled global channels fail.

3 SETTING

Problem and data. We study the Euclidean Traveling Salesman Problem with $N = 100$ nodes drawn i.i.d. uniformly from $[0, 1]^2$. Evaluation uses a fixed held-out set of 1000 instances; the oracle is LKH-3 [Helsgaun, 2017] with mean optimal tour length 7.749371521. All gaps are reported as percent excess over this oracle mean.

Training. Every encoder is trained with POMO policy-gradient RL [Kwon et al., 2020]: REINFORCE with a shared-baseline advantage equal to the reward minus the per-instance mean over 100 POMO rollouts. The decoder (the POMO TSP decoder) is byte-identical across all encoders; only the encoder is swapped. We use Adam with learning rate 1×10^{-4} , weight decay 1×10^{-6} , batch size 64, and 250k gradient steps (500k for the converged extensions). Each run uses a single NVIDIA A100-SXM-64GB on the CINECA Leonardo Booster.

Evaluation protocol. We report two metrics. *Single-trajectory* gap is greedy argmax decoding averaged over the 100 POMO start nodes. *Multistart* gap is the best of the 100 start-node rollouts, with *no* instance augmentation. We deliberately omit the $\times 8$ augmentation used in headline POMO numbers so that the encoder, not the augmentation, is what the comparison measures.

Relation to published POMO numbers. Published POMO reaches $\sim 1\%$ greedy on TSP-100 at a much larger training budget; we reproduced the official released checkpoint at 0.97% greedy (0.149% with $\times 8$ augmentation) under our exact oracle and evaluation set as a seatbelt before any experiment (artifact in the repository). Our attention baseline is therefore not undertrained relative to its budget; it is the same architecture trained at a deliberately smaller, *matched* budget (250k steps of batch 64, $\approx 16\text{M}$ instances), because

Table 1. Encoder configurations, matched in total parameters. The hybrid has the fewest parameters (4.5% below attention).

Encoder	Layers	Serialization	Parameters
Attention	6 ($d=128$, 8 heads)	n/a (permutation-invariant)	1,269,760
Uni-Mamba	10 Mamba-1 blocks	Hilbert / random / sort	1,249,792
BiMamba	5 (fwd+bwd scan, summed)	Hilbert / random	1,248,512
Hybrid	4 BiMamba + 1 attention (after layer 2)	Hilbert	1,213,184

every claim in this paper is a matched-budget comparison between encoders, not a state-of-the-art claim. Absolute gaps here should not be compared against headline POMO numbers.

Parameter matching. The encoders are matched in total model size to within 4.5% (Table 1): attention (6 layers, $d = 128$, 8 heads) totals 1,269,760 parameters; the unidirectional Mamba encoder (10 Mamba-1 blocks, $d = 128$) totals 1,249,792; the bidirectional Mamba (5 layers, each summing a forward and backward scan) totals 1,248,512; and the hybrid (4 bidirectional-Mamba layers with one POMO-native attention layer inserted after layer 2) totals 1,213,184, which is 4.5% *fewer* parameters than attention. The Mamba runs use the mamba-ssm CUDA kernel, gated against a pure-PyTorch reference at $< 10^{-3}$ agreement. Throughput on the A100 is comparable across encoders (attention ~ 5.1 , unidirectional Mamba 5.03–5.04, bidirectional 4.78, hybrid ~ 4.8 it/s).

Serialization. A scan encoder needs a node order. We serialize the nodes, scan, and un-serialize back to the original node order before decoding. The default order is the Hilbert curve; we also study a fresh random permutation per instance and a coordinate (lexicographic) sort as ablation arms.

4 THE KILL AND THE DIAGNOSIS

The vanilla recipe is killed. Before running, we fixed a kill rule: kill the candidate if it is worse than attention by more than 1 percentage point *and* Hilbert clearly beats random ordering. At the matched 250k-step budget the Hilbert-ordered unidirectional Mamba reaches an 8.34% single-trajectory gap and 4.46% multistart on the pre-registered seed (three-seed mean 7.50 ± 1.03 / 3.64 ± 1.09 ; even its best seed trails attention by more than 3 points), versus attention’s 2.95% / 1.91% (Table 2). The first clause of the kill rule is satisfied by a wide margin under every seed; as §5 shows, the second clause resolves in an unanticipated direction. The encoder-swap-as-specified is dead. We now ask *why*, using four diagnostics on the trained Hilbert/uni-Mamba seed-1 checkpoint.

(1) It is not a wiring bug: order reaches the decoder. We swap the inference-time scan order on the trained checkpoint without retraining, on an $n = 200$ subset (hence the small offset from the $n = 1000$ figures elsewhere). The gap moves from 8.38% (Hilbert, the training order) to 33.49% (sort) to 56.38% (random). Feeding the wrong order collapses the model, so the serialize→scan→un-serialize→decode path is intact and strongly order-dependent; the failure is not a plumbing bug.

(2) It is not representation collapse. On 10k node embeddings, the Mamba encoder has effective rank 31.7 (attention: 26.0), mean pairwise cosine similarity 0.116 (0.102), and zero dead dimensions (as does attention). The embeddings are at least as healthy and high-rank as attention’s; the deficit is in *what* they encode, not encoder health.

(3) The receptive field is short and one-directional. Finite-difference sensitivity shows that upstream sensitivity is exactly zero (a node never sees its sequence-successors), and downstream influence decays $\sim 99\%$ within ~ 20 sequence positions. Each node integrates only a short, one-directional window, consistent with the causal-scan limitation of PointMamba [Liang et al., 2024].

(4) The Hilbert prior scatters spatial neighbors. A model-free geometry study shows that at $N=100$ the Hilbert curve places 14.4% of each node’s $k = 5$ nearest 2D neighbors more than 10 sequence positions apart on the scan line (curve-boundary discontinuities). A short causal window cannot reconnect them, whereas all-pairs attention sees them for free. The scatter worsens with scale: the mean line-separation of spatial neighbors grows from 7 to 48 as N grows from 100 to 5000 (fraction >10 apart: 14.6% $\rightarrow$ 22.2% under the scaling study’s estimator, which differs slightly from the 14.4% $k=5$ figure above), while the encoder’s receptive field is fixed (~ 20). This is a geometry of the serialization, not a contribution of ours; it predicts the limitation *worsens* at the larger N where $O(N)$ would otherwise pay off.

5 ORDERING ABLATION AND THE SEED LESSON

The second clause of the kill rule (“Hilbert clearly beats random”) turns out to be false, and the way it turned out is itself a methodological lesson. We trained unidirectional Mamba from scratch under three serializations, identical in every other respect (Figure 1).

The seed-1 inversion. On seed 1, the ordering hierarchy inverted relative to the locality prior: random 6.40% < sort 7.12% < Hilbert 8.34% single-trajectory, with random ahead of Hilbert at every 20k-step checkpoint. Read literally, this refutes the Hilbert-locality premise: spatial-locality serialization does not help this encoder; it hurts.

Refutation by further seeds. Seeds 2–3 do not sustain the strict inversion. Across three seeds, Hilbert spans 6.36–8.34 (mean 7.50 ± 1.03) and random spans 6.40–7.45 (mean 6.90 ± 0.53); the ranges overlap, and Hilbert seed 2 in fact beats every random seed. The strict “random < sort < Hilbert whole-curve” claim was a seed-1 artifact and must not be stated unqualified.

The surviving claim. What survives is weaker and, we think, more useful: *a locality prior confers no measurable benefit over random ordering for a causal scan, and roughly doubles run-to-run variance* (σ 1.03 for Hilbert vs 0.53 for random). This is closer to Point Cloud Mamba’s “simple $\approx$ curve orders” [Zhang et al., 2025a] than to a strict inversion. The pre-registered expectation that Hilbert would clearly beat random finds no support.

Causality-dependence. The value of a locality prior depends on scan causality. With the bidirectional scan of §6, random ordering instead *hurts*: BiMamba/random reaches 7.61% versus BiMamba/Hilbert 5.57%. Mechanistically, once both scan directions are available, Hilbert locality becomes exploitable structure and random ordering destroys it; for a causal scan it is a crutch the short window overfits. The one-line takeaway is that the “best” serialization is not a property of the geometry alone but of the geometry *and* the scan’s directionality.

We frame this as a methodological finding as much as an empirical one: a single-seed ordering curve, however cleanly separated, was enough to state an inverted hierarchy that two further seeds dissolved. The pre-registered decision rule survived the surprise only because the seed budget was part of the plan.

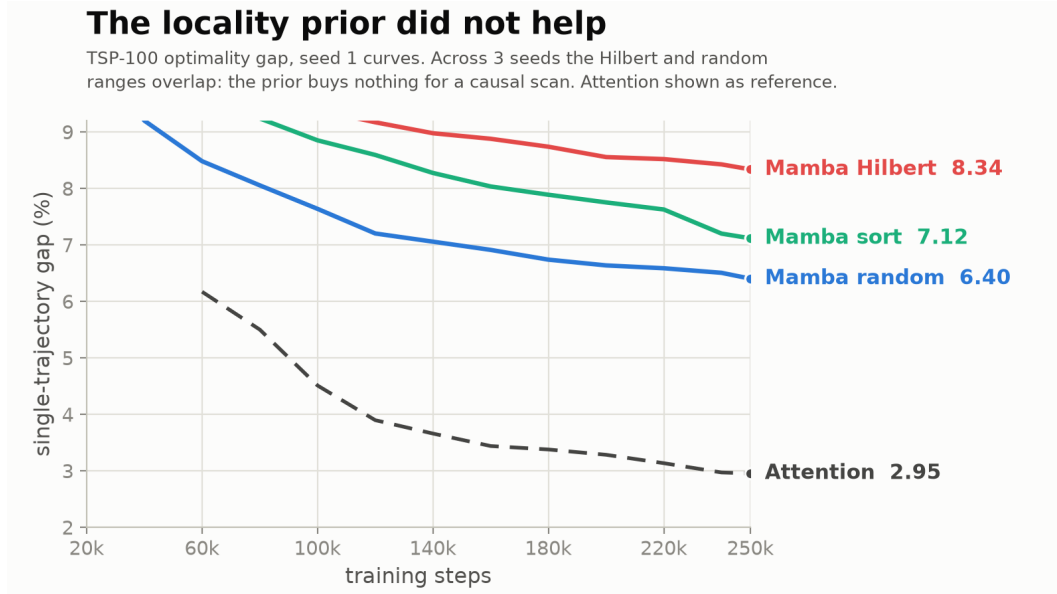


Figure 1. Ordering ablation for the unidirectional Mamba encoder (single-trajectory gap vs training step). On seed 1 the hierarchy separates as random < sort < Hilbert at every checkpoint; seeds 2–3 (§5) show the Hilbert and random distributions overlap. The surviving claim is that a locality prior gives no benefit for a causal scan and roughly doubles seed variance.

6 REPAIR PATH

The diagnosis points at a global-context deficit over a short causal window. We test three repairs in increasing order of what they change, and let the pre-registered decision rules decide.

6.1 Global channels fail

The cheapest repair adds a global channel to the encoder output, $H \rightarrow H + G(H)$, leaving the decoder untouched. We pre-registered an A/B rule before running. Two variants: a mean-pool channel (+1.32% params) and a structured 10-segment attention-over-summaries channel (+2.62% params). Neither helps: mean-pool reaches 7.86% and structured reaches 8.38%, versus the uni/Hilbert baseline of 8.34%. Both moves are within the Hilbert seed noise ($\sigma \approx 1.0$). The receptive-field deficit is not repaired by cheap global summaries under RL.

6.2 Bidirectionality: parity and a tenfold variance reduction

A param-matched bidirectional Mamba (5 layers, each summing a forward and backward scan; 1,248,512 params) directly attacks the one-directional half of the diagnosis. Its single-trajectory gap of 5.57 ± 0.11 closes roughly half of the uni-Mamba gap. The more striking result is on multistart: 1.94 ± 0.09 across three seeds, on par with attention’s budget-matched 1.91 (a single seed; with $n=3$ vs $n=1$ we report ranges rather than tests). Under multistart decoding the bidirectional encoder supports tours as good as attention’s, at $O(N)$. Two caveats keep this at “parity” rather than “win”: it is a matched-250k comparison, and the residual deficit is concentrated in single-trajectory greedy decoding. Bidirectionality also stabilizes training: the seed variance (± 0.11 single-trajectory) is an order of magnitude tighter than uni-Mamba/Hilbert’s (± 1.03).

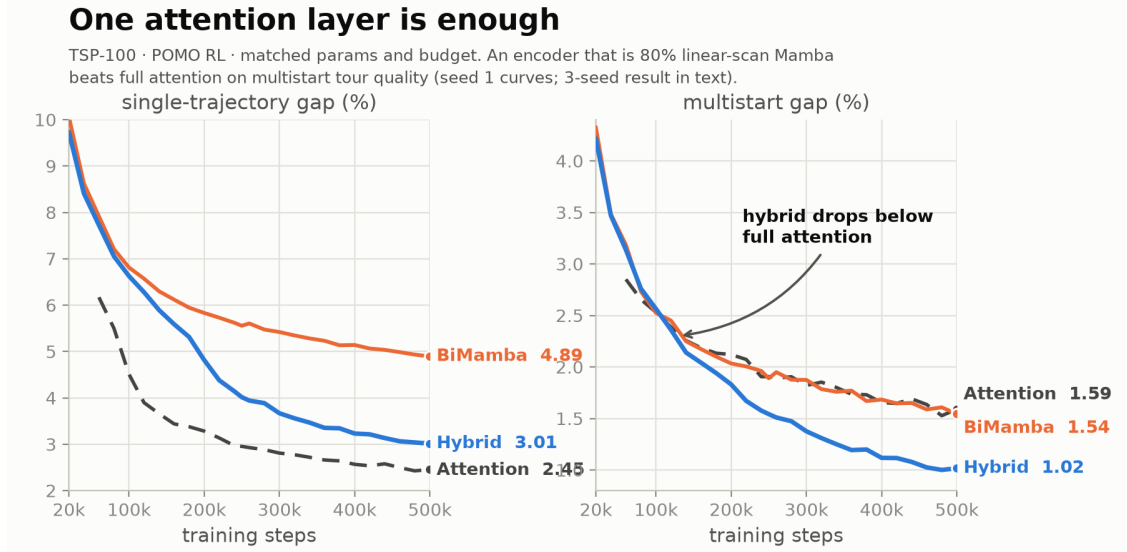


Figure 2. Hybrid encoder (4 bidirectional-Mamba layers + one attention layer, 4.5% fewer parameters than attention) versus the all-attention baseline. At the converged 500k budget the hybrid beats attention on multistart tour quality (§6), while attention remains ahead on single-trajectory greedy decoding. All gaps are percent over the LKH-3 oracle.

6.3 Hybrid: one attention layer beats attention on multistart

Where ten pooled-summary channels failed, one exact attention layer succeeds. The hybrid inserts a single POMO-native attention EncoderLayer after layer 2 of a 4-layer bidirectional-Mamba stack, inside the serialized stream, at 1,213,184 parameters (4.5% fewer than attention). At 250k it reaches 4.37 ± 0.33 single-trajectory and 1.63 ± 0.10 multistart across three seeds, and every hybrid seed’s multistart (1.54/1.61/1.74) beats attention’s budget-matched 1.91 (Table 2).

At the converged 500k budget (Table 3, Figure 2) the hybrid decisively beats attention on multistart. Across three seeds on each side, the hybrid reaches 3.08 ± 0.06 single / 1.14 ± 0.19 multistart versus attention’s 2.52 ± 0.07 / 1.61 ± 0.02 , and the *worst* hybrid seed’s multistart (1.365) is still below the best attention figure (1.589), well outside the ± 0.05 multistart eval-noise band (estimated from the fluctuation of the final three evaluations on the flat tails of the 500k curves). Wherever this paper says “beats,” it means exactly this scoped multistart claim, with the noise band and seed range stated. On single-trajectory, attention remains ahead (~ 2.45 – 2.78 vs the hybrid’s 3.08 ± 0.06). We are explicit that the interleaving pattern is prior art [Lieber et al., 2024, Glorioso et al., 2024]; the finding is that sparse *exact* attention, not pooled global averages, is the repair that works under RL routing.

6.4 Best-of- k serialization ensembling

A scan encoder has a degree of freedom attention lacks: the serialization. At inference on random-trained *seed-1* checkpoints we draw $k = 8$ random orders and keep the per-instance best tour. This lowers uni-Mamba/random multistart from 3.33% to 2.72% (average-start 6.42% $\rightarrow$ 5.01%) and BiMamba/random from 3.31% to 2.70% (the $k=1$ figures reproduce the respective seed-1 training-time evals to within order-resampling noise; the three-seed means in Table 2 differ slightly). Attention has no ordering degree of freedom, so this is a diversity knob unique to scan encoders. We report it as a supporting observation; its natural host is a strong random-trained model, which (per §5) does not exist in the bidirectional regime.

7 DISCUSSION AND LIMITATIONS

Table 2. Main results at the matched 250k-step budget. Optimality gap (% over LKH-3), single-trajectory with multistart in parentheses; mean $\pm$ std over the noted seeds. Attention and multistart use no instance augmentation. Every hybrid seed’s multistart is below attention’s 1.91.

Encoder / order	Seeds	Single-traj (%)	Multistart (%)
Attention	s1	2.95	1.91
Uni-Mamba / Hilbert	s1,s2,s3	7.50 ± 1.03	3.64 ± 1.09
Uni-Mamba / random	s1,s2,s3	6.90 ± 0.53	3.28 ± 0.14
Uni-Mamba / sort	s1	7.12	3.64
GC mean-pool (+1.32%)	s1	7.86	4.13
GC structured (+2.62%)	s1	8.38	4.51
BiMamba / Hilbert	s1,s2,s3	5.57 ± 0.11	1.94 ± 0.09
BiMamba / random	s1	7.61	3.30
Hybrid	s1,s2,s3	4.37 ± 0.33	1.63 ± 0.10

N=100 only; the $O(N)$ payoff is untested. Every number here is at $N=100$, where attention’s $O(N^2)$ is cheap. The motivation for a linear-scan encoder is large N , and our own geometry study (§4) predicts the receptive-field deficit *worsens* with scale. Whether the hybrid’s multistart advantage survives, and whether $O(N)$ actually pays, at $N \geq 1000$ is the next gate, not a claim of this paper.

Single-trajectory still favors attention. Our win is a multistart result. On single-trajectory greedy decoding attention remains ahead at every budget we ran. A reader who cares about greedy decoding should read this paper as parity-at-best for the state-space encoders.

The hybrid interleaving is prior art. Mixing attention layers into an SSM stack is established in language models [Lieber et al., 2024, Glorioso et al., 2024] and local/global aggregation in point clouds [Zhang et al., 2025b]. Our contribution is the RL-routing evidence and the contrast we can draw because we ran both: one *exact* attention layer works where two pooled global-channel variants did not. That contrast, not the architecture, is the result.

Fairness caveats. The 500k multistart comparison is three-seed on both sides: hybrid 1.14 ± 0.19 versus attention 1.61 ± 0.02 , with the worst hybrid seed (1.365) below the best attention seed (1.589). Attention at 250k remains a single seed; there and wherever seed counts are unequal we report ranges and worst-vs-best comparisons rather than significance tests. Multistart differences below ~ 0.05 are within eval noise and should not be read as wins. Finally, the throughput table shows no speed advantage for the scan encoders at $N=100$ (bidirectional 4.78 vs attention ~ 5.1 it/s): at this size the paper’s result is about *quality at matched cost*, and the efficiency motivation rests entirely on the untested $N \geq 1000$ regime.

8 CONCLUSION

We set out to test whether a linear-time state-space encoder can route as well as attention on TSP-100 under the reinforcement learning that NCO actually uses, with the decoder and parameter budget held fixed. The

Table 3. Converged results at 500k steps (curves flat). Optimality gap (% over LKH-3), single-trajectory (multistart). All rows are final 500k numbers; the multistart eval-noise band is ± 0.05 . The hybrid-attention multistart separation is three-seed on both sides: worst hybrid 1.365 vs best attention 1.589.

Encoder	Seeds	Single-traj (%)	Multistart (%)
Attention	s1	2.454	1.589
Attention	s2	2.597	1.632
Attention	s3	2.503	1.621
Attention	s1,s2,s3 (mean)	2.52 ± 0.07	1.61 ± 0.02
BiMamba / Hilbert	s1	4.893	1.544
Hybrid	s1	3.013	1.016
Hybrid	s2	3.098	1.051
Hybrid	s3	3.136	1.365
Hybrid	s1,s2,s3 (mean)	3.08 ± 0.06	1.14 ± 0.19

vanilla Hilbert+Mamba recipe is killed, and the failure is a short one-directional receptive field, not a bug or collapsed representations. Following the diagnosis, pooled global channels do not help, but bidirectionality reaches multistart parity with attention while cutting seed variance tenfold, and a hybrid with a single exact attention layer, 4.5% smaller than the attention encoder, beats attention on converged multistart tour quality. Along the way we found that the value of a serialization prior depends on scan causality, and that a clean single-seed ordering curve was enough to state an inversion two further seeds dissolved. All of this is scoped to $N=100$; whether the linear-scan payoff materializes at larger N is the open question we hand to the next gate.¹

ACKNOWLEDGMENTS

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REPRODUCIBILITY

All runs share a byte-identical POMO decoder, optimizer (Adam, lr 1×10^{-4} , weight decay 1×10^{-6}), batch size 64, and gradient budget (250k, 500k for extensions), on a single A100; only the encoder is swapped. Evaluation uses a fixed 1000-instance held-out set with the LKH-3 oracle mean 7.749371521. The kill rule, the BiMamba three-bucket reading, and the global-channel A/B rule were each written and committed *before* the corresponding runs. Code, seeds, curves, and the pre-registered rules are released at <https://github.com/Umaraslam66/serpentine> and archived at DOI 10.5281/zenodo.21517264. The mamba-ssm CUDA kernel was gated against a pure-PyTorch reference at $< 10^{-3}$ agreement.

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¹Code, seeds, the fixed evaluation set, and the pre-registered decision rules are at <https://github.com/Umaraslam66/serpentine>.

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